



Shiva, the Hindu god who oversees health, is traditionally thought to live under the bael tree, an important medicinal plant in India.

on. Such treatment is based on a wealth of acutely observed plant lore and knowledge, handed down in an oral tradition from generation to generation.

The Development of Medicinal Lore

It is generally recognized that our ancestors had a wide range of medicinal plants at their disposal, and that they likewise possessed a profound understanding of plants' healing powers. In fact, up until the 20th century, every village and rural community had a wealth of herbal folklore. Tried and tested local plants were picked for a range of common health problems and taken as teas, applied as lotions, or even mixed with lard and rubbed in as an ointment.

But what were the origins of this herbal expertise? There are no definitive answers. Clearly, acute observation coupled with trial and error has played a predominant role. Human societies have had many thousands of years to observe the effects—both good and bad—of eating a particular root, leaf, or berry. Watching the behavior of animals after they have eaten or rubbed against certain plants has also added to medicinal lore. If one watches sheep or cattle, they almost unerringly steer a path past poisonous plants such as ragwort

(*Senecio jacobaea*) or oleander (*Nerium oleander*). Over and over and above such close observation, some people have speculated that human beings, like grazing animals, have an instinct that recognizes poisonous as opposed to medicinal plants.

Ancient Civilizations

As civilizations grew from 3000 BCE onward in Egypt, the Middle East, India, and China, so the use of herbs became more sophisticated, and the first written accounts of medicinal plants were made. The Egyptian Ebers papyrus of c. 1500 BCE is the earliest surviving example. It lists dozens of medicinal plants, their uses, and related spells and incantations. The herbs include myrrh (*Commiphora molmol*, p. 85), castor oil (*Ricinus communis*, p. 262), and garlic (*Allium sativum*, p. 59).

In India, the *Vedas*, epic poems written c. 1500 BCE, also contain rich material on the herbal lore of that time. The *Vedas* were followed in about 400 BCE by the *Charaka Samhita*, written by the physician Charaka. This medical treatise includes details of around 350 herbal medicines. Among them are visnaga (*Ammi visnaga*, p. 62), an herb of Middle Eastern origin that has recently proven effective in the treatment of asthma, and gotu kola (*Centella asiatica*, p. 76), which has long been used to treat leprosy.



Castor oil plant has been used medicinally for about 4000 years.

Medicine Breaks from its Mystical Origins

By about 500 BCE in developed cultures, medicine began to separate from the magical and spiritual world. Hippocrates (460–377 BCE), the Greek “father of medicine,” considered illness to be a natural rather than a supernatural phenomenon, and he felt that medicine should be given without ritual ceremonies or magic.

In the earliest Chinese medical text, the *Yellow Emperor's Classic of Internal Medicine* written in the 1st century BCE, the emphasis on rational medicine is equally clear: “In treating illness, it is necessary to examine the entire context, scrutinize the



The bump in the right cheek of this Peruvian figurine may represent coca, taken in that country to increase endurance.